Is photoperiod the dominant or least important cue for woody plant spring phenology?

Woody plant leafout and flowering each spring is strongly driven by warm spring temperatures, and thus has shifted days to weeks earlier with anthropogenic climate change. But how long this advance can continue depends on the relative importance of the two other major environmental cues—chilling (via the accumulation of cool winter temperatures) and photoperiod. Over a decade of focused research (e.g., Laube *et al.*, 2014; Ettinger *et al.*, 2020) has found chilling produces the largest shifts in spring phenology, often 3-5X larger than the response to photoperiod (Morales-Castilla *et al.*, 2024). But this research generally tests photoperiod responses in only one way (Fig. 1) that is disconnected from the molecular concept of critical photoperiod. At the same time, new studies suggest a dominating role of solstice mid-season phenology (Zohner *et al.*, 2023; Journé *et al.*, 2024), and a remarkably stable photoperiod response across most species (Fig. 2). Together these new results suggest an important and poorly understood role of photoperiod that may not be well tested by current methods. Building a mechanistic predictive model of spring phenology thus requires re-examining how much we actually know currently and developing new approaches.

Here, I want to explore:

1. How do current findings in woody plant spring phenology experiments compare to our understanding of photoperiod from the molecular literature? Most studies manipulate photoperiod alongside forcing and chilling temperatures usually by using two different constant photoperiods with only small responses (Fig. 1).

2. What experiments could help better understand mechanistically how photoperiod affects spring phenology?

We have data on many woody plant spring phenology experiments and can explore:

1. Differences in photoperiod responses for budburst versus flowering (since most molecular research is on flowering, right?).

2. Budburst % versus timing responses to photoperiod... would need to back up in the OSPREE database some to do this, but would be good to also show for a subset of species (Fagus and Betula perhaps?)

3. Whether any studies do light interruption (I have not found any in our database of woody plant spring phenology, but I saw some for strawberry, which was included by accident). That's lieten97 ... but I think we did a poor job of documenting this paper, maybe it is: `https://www.actahort.org/books/439/439_105.htm`

4. Compare findings we see from molecular experiments such as those that Stacey Harmer pointed out where the response is considered not important, but is actually just small compared to the larger responses seen usually to critical photoperiod.

Thinking in 2026.... Other stuff we could add...

1. Dan's stuff on male vs female flowers

2. Dan's stuff on how photoperiod and thermoperiodicity often covaried

3. Korner work on leafout is photoperiod (cell elongation/photosynthesis) while budburst is a trigger (more like the flowering literature)
4. Deirdre's work on traits (photoperiod matters)
5. Photoperiod and budset (but Rob Guy actually did photoeperiod interruption experiments)
6. Better models of forcing (log-log or threshold ones) and what we find in photoperiod effects then...
7. Probably need to give a REALLY concrete example of why knowing the mechanism matters.

Be also sure to review what we say in Nacho's 2024 paper about photoperiod and phylogeny as it's good. We can just repeat some of this....

* write paper, then do any additional analyses; send to Dan, then ask Stacey Harmer to join it after quick draft? Maybe see if Justin wants to join, since he is a molecular expert!

Outline

1. Introduction
   (a) Phenology matters
   (b) But does photoperiod?
   (c) And something about climate change goes here..
2. Current well-established models of photoperiod
   (a) Hourglass
   (b) Whatever you call the other ones ...
3. Evidence in woody plant phenology (this section needs some subsections...)
   (a) Lots about critical photoperiods and stuff...
   (b) But studies never test for critical photoperiods
   (c) And if you look at the data, it's not so dramatic (Walde, OSPREE results)
   (d) But it's there! Nacho's OSPREE paper.... bridge evolution back to molecular ...
4. New models of photoperiod
   (a) Photosynthesis matters
   (b) Imaizumi et al. 2025 (who is at the UW!) – overviews recent 'more natural' experiments that have turned up weird stuff → this is where phenological studies come in!
   (c) How do we bridge these fields!?
5. Paths forward
   (a) The molecular people are starting to use natural experiments, they should keep doing that ...
   (b) Us climate change phenology people may need some night interruption experiments
   (c) Look more at flower vs leafout?
   (d) Should we maybe focus on some model species? Should Fagus be on (maybe no, but maybe a Quercus species?) and a Populus one?

References

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Figures

..//..//cherries/data/walde_etal/figures/waldehighchillphoto.pdf

Figure 1: Results from Walde *et al.* 2023 for budburst timing across species under high chilling and two photoperiods: 8 and 16 hours. This type of study is the most common type done in woody plant phenology—it uses cuttings of dormant buds exposed to chilling (cool temperatures) followed by different combinations of warm temperatures and photoperiods (this study is unique though in covering a wide range of forcing temperatures).

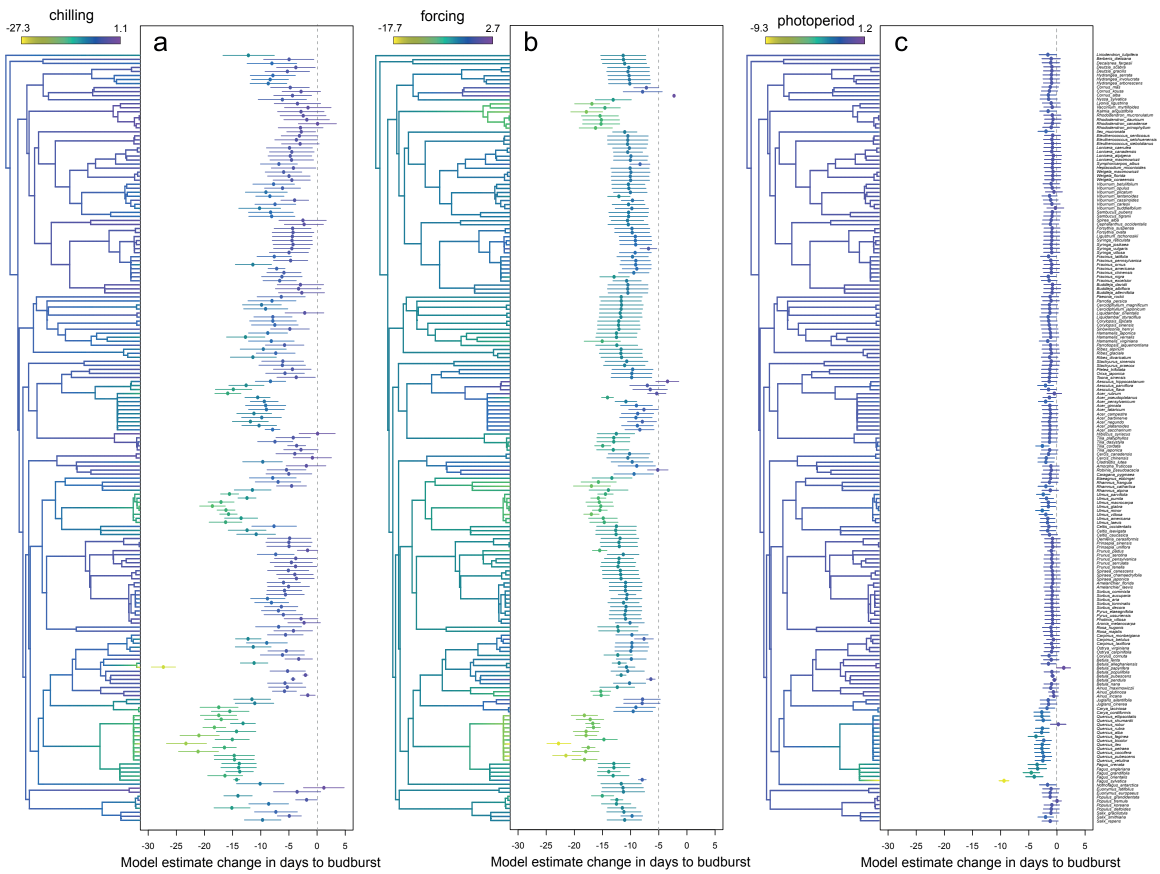

Figure 2: Estimated chilling, forcing and photoperiod responses (in units standardized by the mean and SD for each response variable) from 191 woody angiosperm species show most species shift earlier two days per standardized photoperiod unit, except for *Fagus* species with the well-studied *Fagus sylvatica* showing the strongest response. Figure reproduced from Morales-Castilla *et al.* (2024).